Lecture 1

Special thanks to Dr. Johann Gagnon-Bartsch

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Today's Lecture

- Reading Summary – Chapter 1
- Statistics and the Scientific Method
- Types of Studies
  - Experiments
    - The Salk Vaccine Trial
    - Randomization

Read Chapters 1 and 2 for next lecture.

Reading Summary

Chapter 1.1: The Salk Vaccine Field Trial

- Treatment
- Outcome (response)
- Treatment group, control group
- Method of comparison
- Confounding
- Randomization (two main benefits)
- Placebo
- Double-blind

Chapter 1.2: The Protcaval Shunt

- Controls matter!
- Randomization matters!

Chapter 1.3: Historical Controls

- Historical controls
- Contemporaneous controls

Statistics and the Scientific Method

Statistics and the Scientific Method

- Scientific Method
  1. Formulate a hypothesis.
  2. Collect data to test the hypothesis.
  3. Interpret data to determine consistency of the data with the hypothesis.
  4. Repeat steps 1-3 as needed.
- A **study** implements the scientific method.
- Statistics facilitates step 3.
- Statistics helps to determine the data collection process and the data collected in step 2.

Types of Studies

Types of Studies

- Experiments
- Natural Experiments
- Observational Studies

Experiments

- In an experiment, the **researcher** has more power over the data collection process and the data collected.

The Salk Vaccine Trial

Polio

- Polio is an infectious disease that can result in paralysis, muscle weakening, and death.
- Polio epidemics in the early 1900's claimed hundreds of thousands of victims, especially children.
- Jonas Salk developed a polio vaccine in the early 1950's.
  - The vaccine was safe in laboratory trials.
  - The vaccine caused polio antibody production in cell cultures.
- How did Salk determine that the vaccine will prevent polio in people?

Some Initial Thoughts

- Why not give everyone the vaccine and calculate the number and percentage of people who get polio?
  - It may be that the people who didn't get polio wouldn't have gotten polio even if they had not gotten the vaccine.
- The study needed to have a **control group** of people who didn't get the vaccine.
- The vaccine was the **treatment** and the people who got the vaccine were the **treatment group**.
- Why not compare the percentage of vaccinated people who got polio to the percentages from previous years?
  - It is usually important that treatment be **contemporaneous** to (at the same time as) control.

Additional Thoughts

- The treatment and control groups should be contemporaneous.
- How should researchers determine the treatment and control groups?
  - Perhaps doctors should give the vaccine to the people who would be more vulnerable to polio.
  - What would be the problem with this study design?
  - Should doctors be choosing the treatment and control groups at all?

Bias

- The results of the study should approximate the true effectiveness of the vaccine.
- **Bias** is the difference between the results of the study and the true effectiveness.
- Study design should account for and limit bias to the extent possible.

Randomization

- It is important that the treatment and control groups be as similar as possible in order to reduce bias.
- Having the doctors choose the treatment and control groups is likely to lead to differences between the treatment and control groups.
- **Randomization** is an important tool to create similar treatment and control groups.
- Randomization ensures that chance enters the study in a planned, quantifiable way.
- It also allows for blinding.

Why Not Randomize?

- A few possible reasons:
  - Sometimes possible to get more comparable groups by hand (especially with small experiments)
  - May be less challenging logistically

Blinding

- The belief that you are receiving a treatment, the **placebo effect**, is powerful.
- It is important that the subject is **blind** to (does not know) their treatment assignment.
- It is also important that anyone interacting with the subject is blind to the treatment assignment.
- A **double-blinded** study is one where neither the subjects nor the researchers know the treatment assignment.

The Salk Vaccine Trial

The Salk Vaccine Trial was a controlled, randomized, double-blinded experiment.

Results

	Size	Rate
Treatment	200,000	28
Control	200,000	71
No Consent	350,000	46

Rate is cases per 100,000.

Results Evaluation

- There appears to be a large reduction in the polio rate between treatment and control.
- How confident are we that this result can't be explained by chance?
  - This will be subject of discussion near the end of the semester.